

WASP-2b

R-1481

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_210715 · created 2026-08-01T12:56:48+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459410.8198 +/- 0.0065 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.048
Transit depth (Rp/Rs)^2	2.53 +/- 1.51 [%]
Orbital Inclination (inc)	89.08 +/- 2.83
Airmass coefficient 1 (a1)	1.1 +/- 0.36
Airmass coefficient 2 (a2)	-0.06 +/- 0.52
Scatter in the residuals of the lightcurve fit is	32.97 %
Best Comparison Star	#1 - [49, 228]
Optimal Aperture	1.8
Optimal Annulus	5.49
Transit Duration (day)	0.09 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.159 ± 0.048 vs 0.1326 (deviation 0.55σ)
Duration fit vs published	0.09 d vs 0.0737 d (deviation 1.35σ)
Mid-time O–C	13.7 min
Depth SNR	3.3
Residual scatter	32.97 %

Light curve

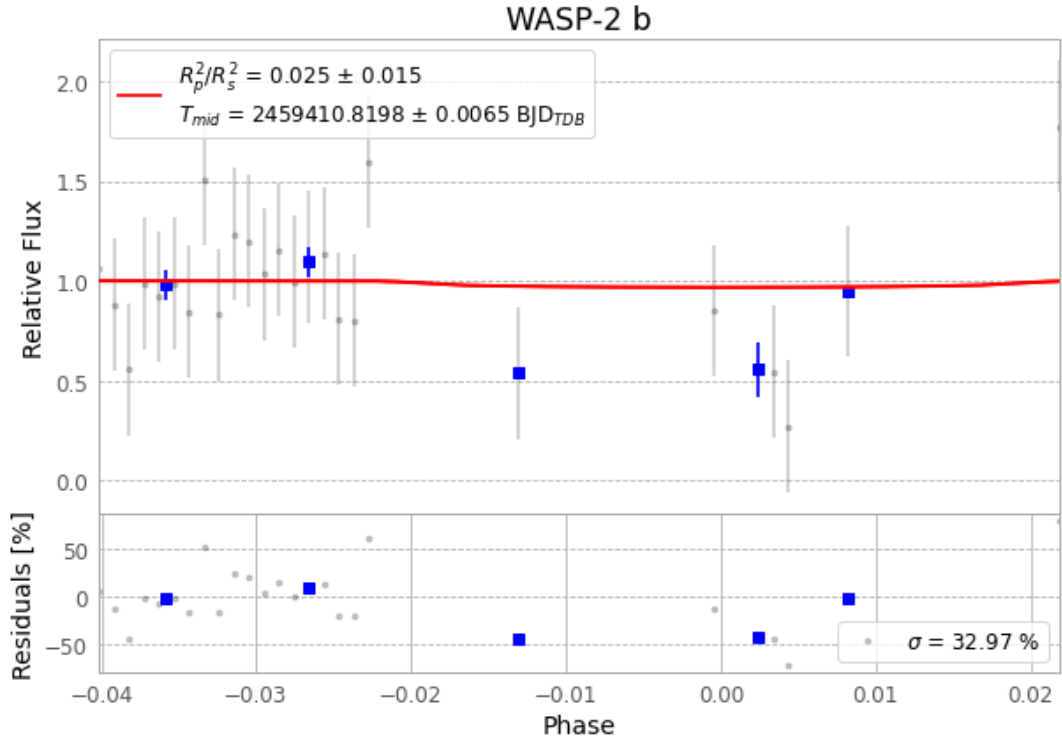


Figure 1 — FinalLightCurve_WASP-2 b_2021-07-14.png (SHA-256 4f15dd15e45fc0da...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [359, 142], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dd0dc3a878562fed...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-2210715052712.FITS → WASP-2210715091212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-210715.log (f9653e622a42...), FinalLightCurve_WASP-2 b_2021-07-14.png (4f15dd15e45f...), FinalLightCurve_WASP-2 b_2021-07-14.csv (791f0b2aeb4c...)

Compute resources

Wall time 230.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 12:56Z.